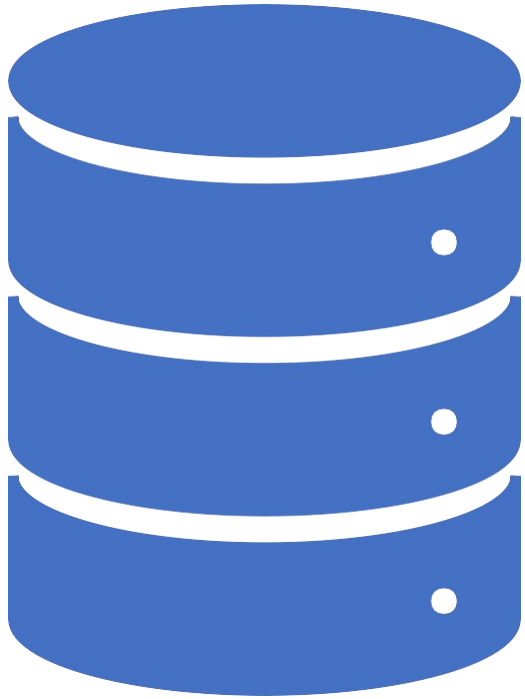
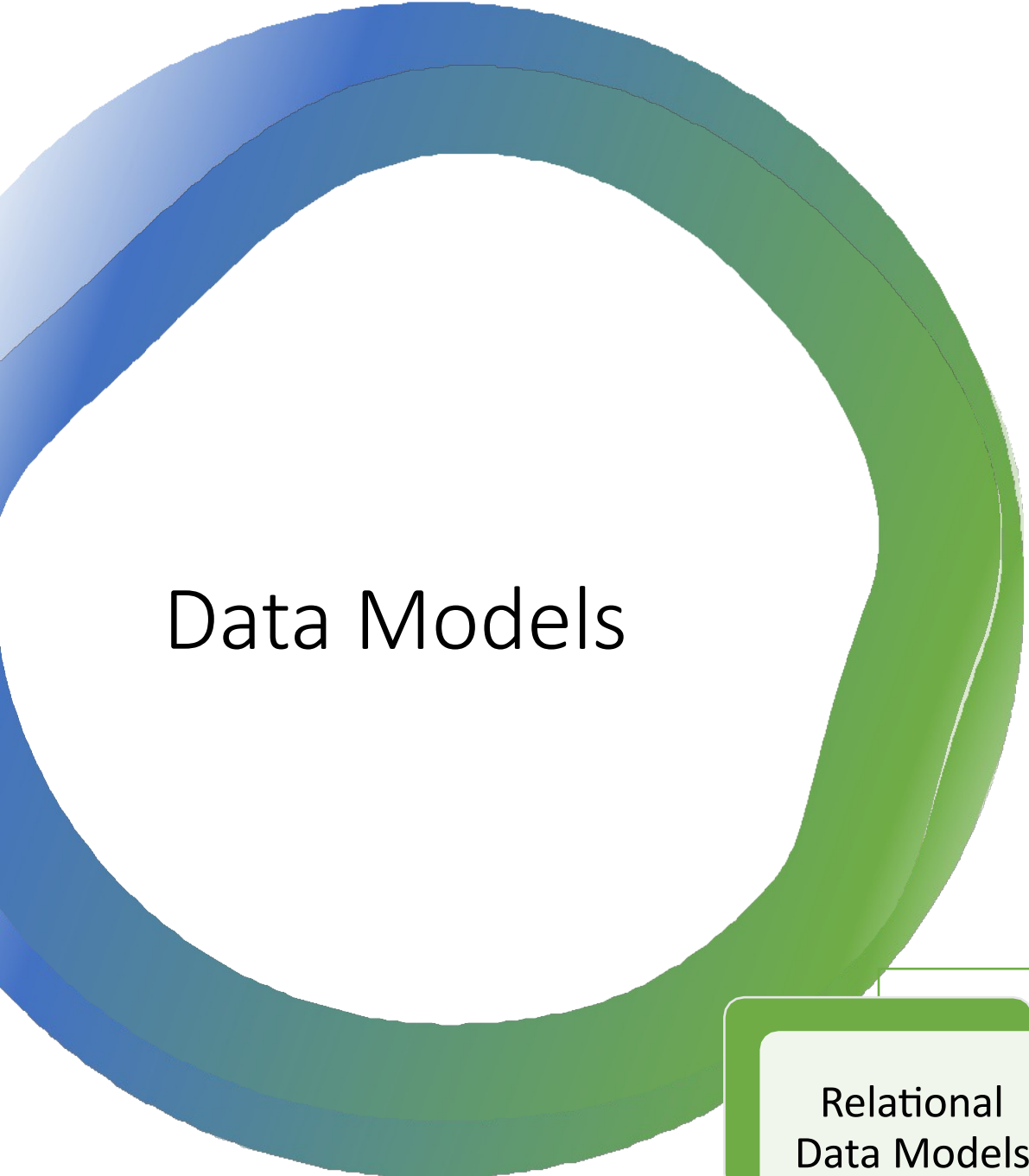




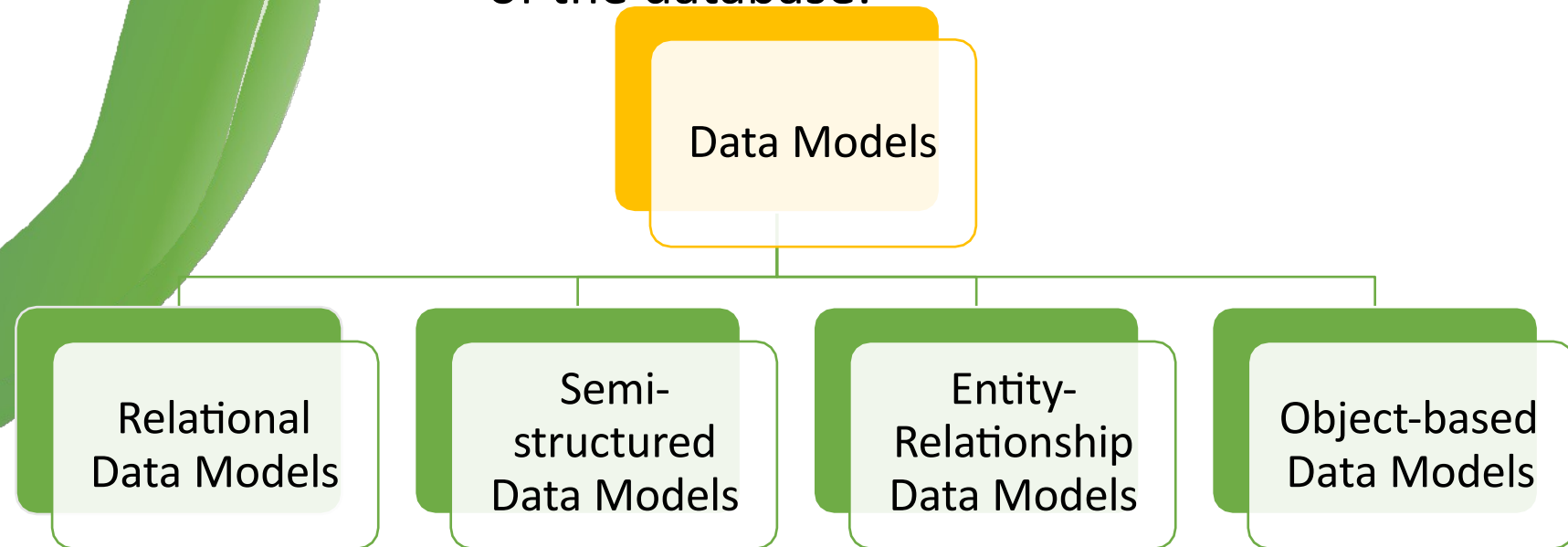
Database Systems

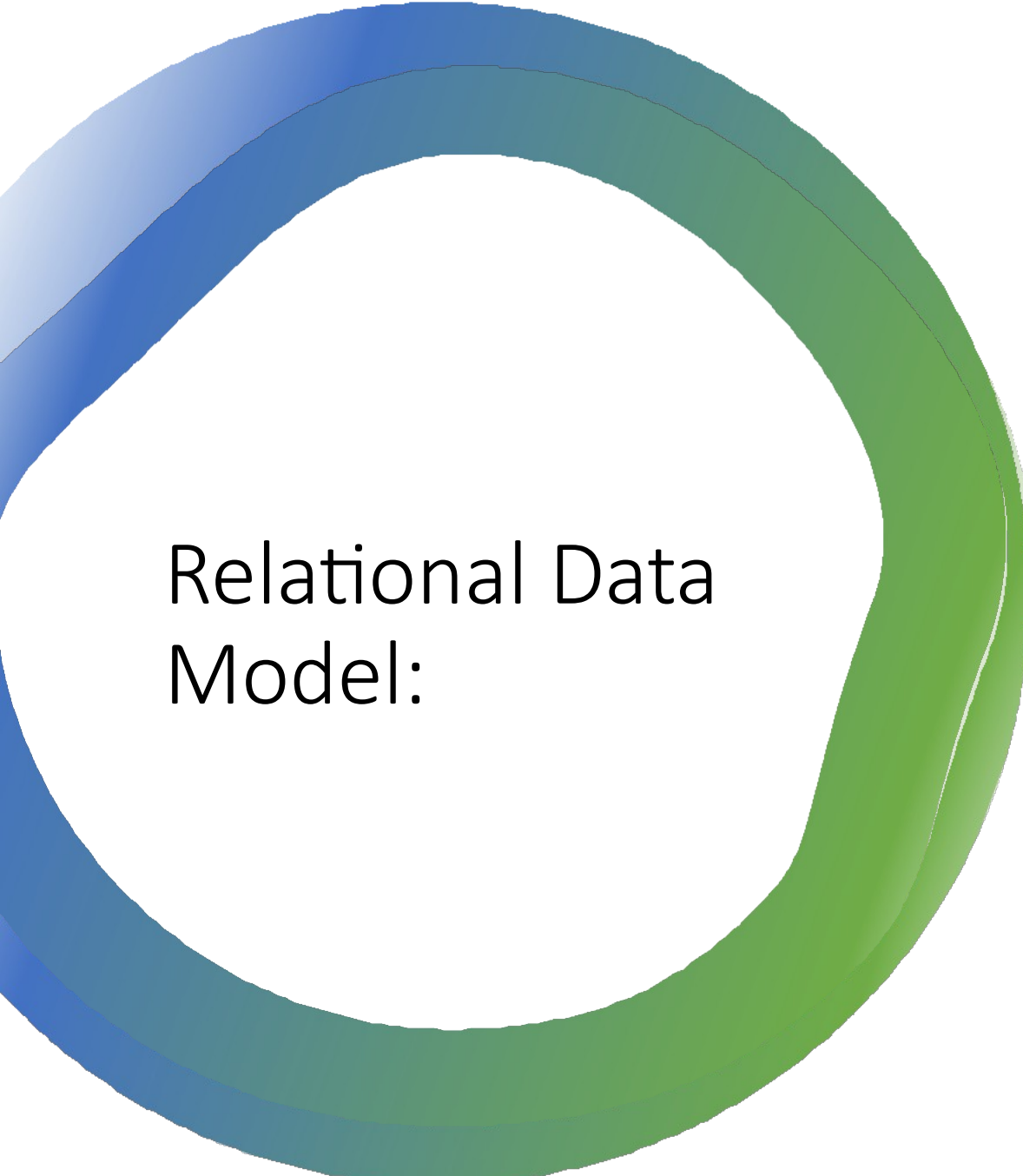
*[Data Models, Schema and
Data Independence]*



Data Models

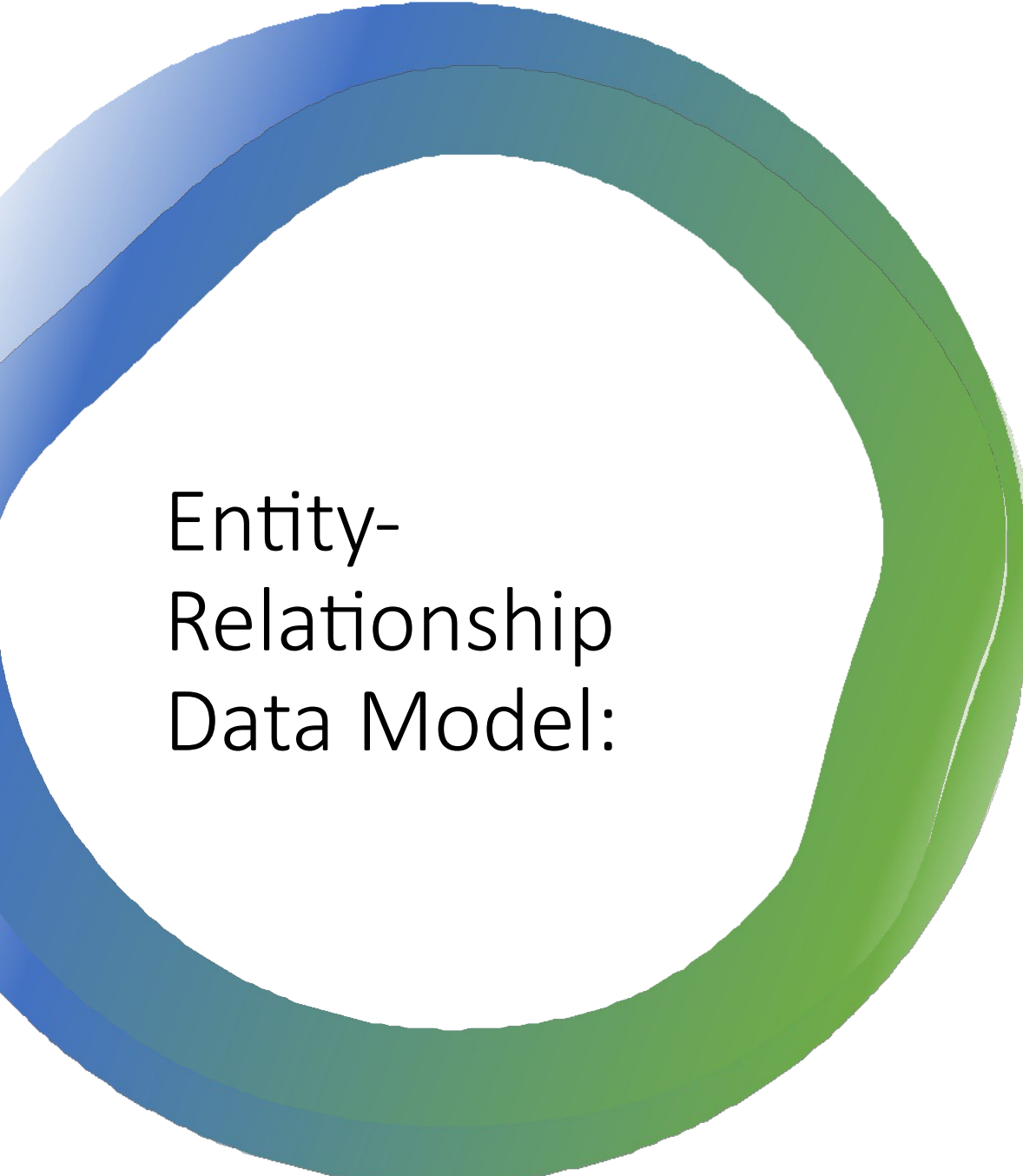
- Data Model is the modeling of the data description, data semantics, and consistency constraints of the data.
- It provides the conceptual tools for describing the design of a database at each level of data abstraction.
- Therefore, there are following four data models used for understanding the structure of the database:





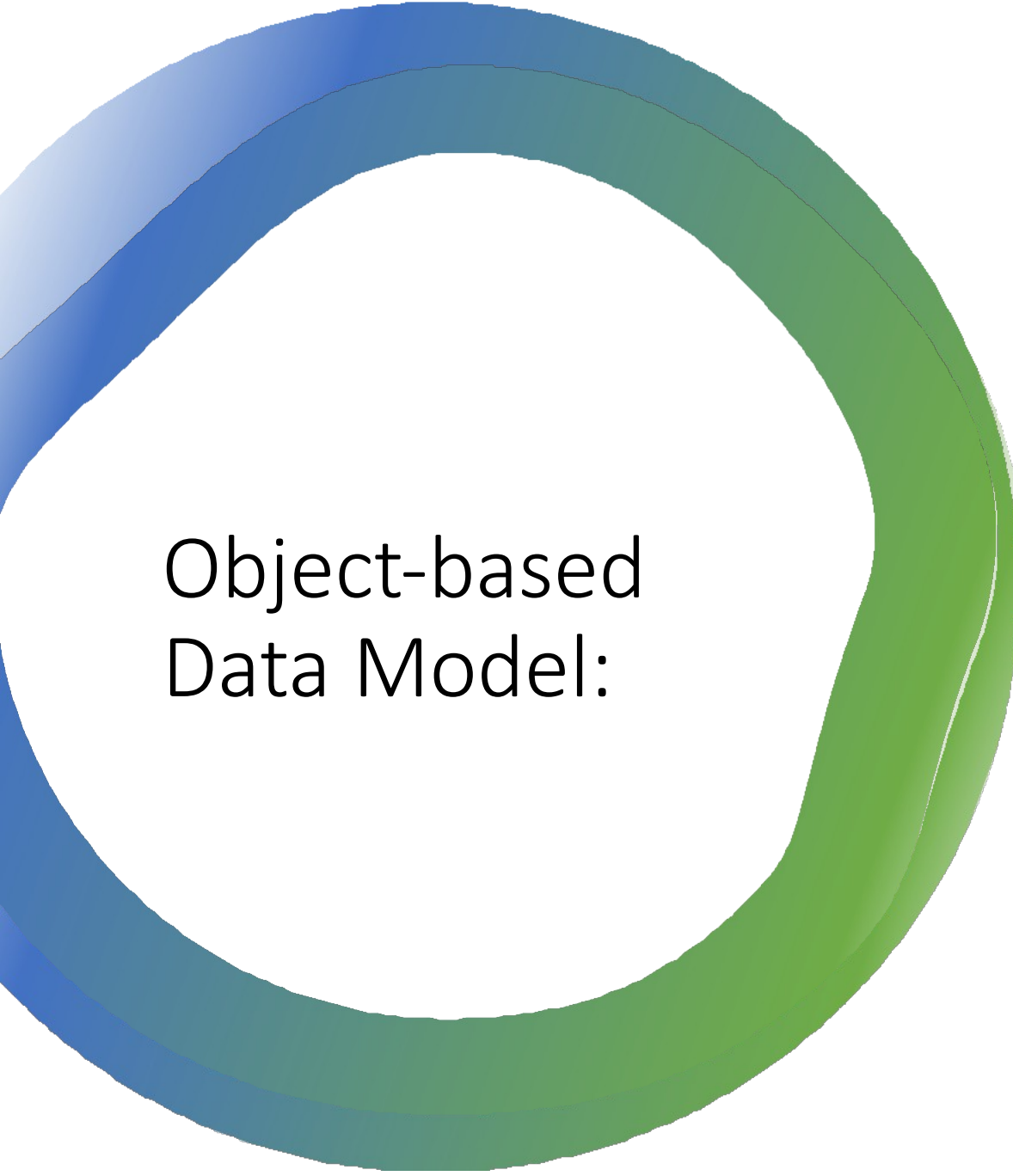
Relational Data Model:

- This type of model designs the data in the form of rows and columns within a table.
- Thus, a relational model uses tables for representing data and in-between relationships.
- Tables are also called ***relations***. This model was initially described by Edgar F. Codd, in 1969.
- The relational data model is the widely used model which is primarily used by commercial data processing applications.



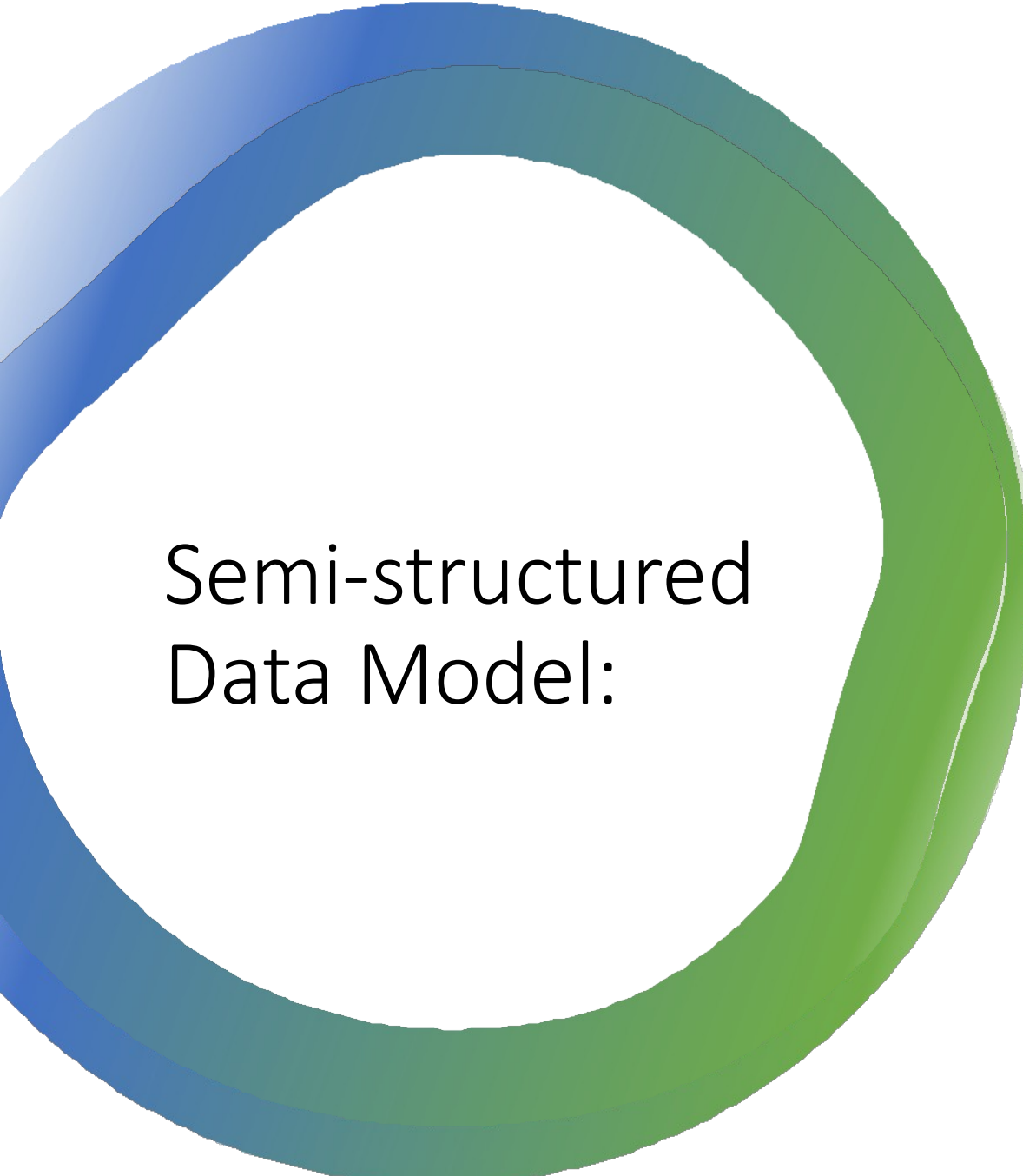
Entity- Relationship Data Model:

- An ER model is the logical representation of data as objects and relationships among them.
- These objects are known as **entities**, and relationship is an association among these entities.
- This model was designed by Peter Chen and published in 1976 papers.
- It was widely used in database designing. A set of attributes describe the entities.
 - For example, student_name, student_id describes the 'student' entity.
- A set of the same type of entities is known as an '**Entity set**', and the set of the same type of relationships is known as '**relationship set**'.



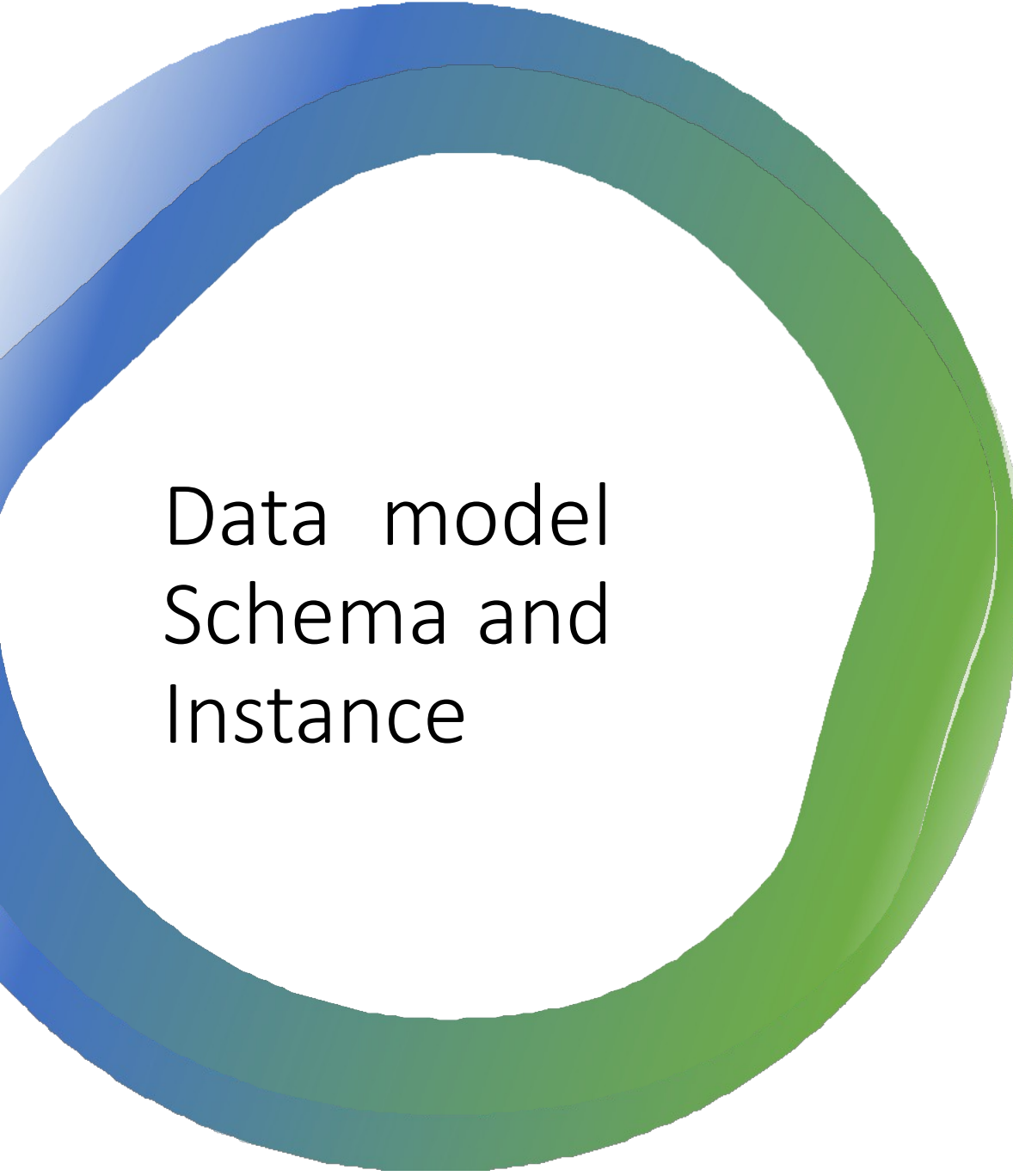
Object-based Data Model:

- An extension of the ER model with notions of functions, encapsulation, and object identity, as well.
- This model supports a rich type system that includes structured and collection types.
- Thus, in 1980s, various database systems following the object-oriented approach were developed.
- Here, the objects are nothing but the data carrying its properties.



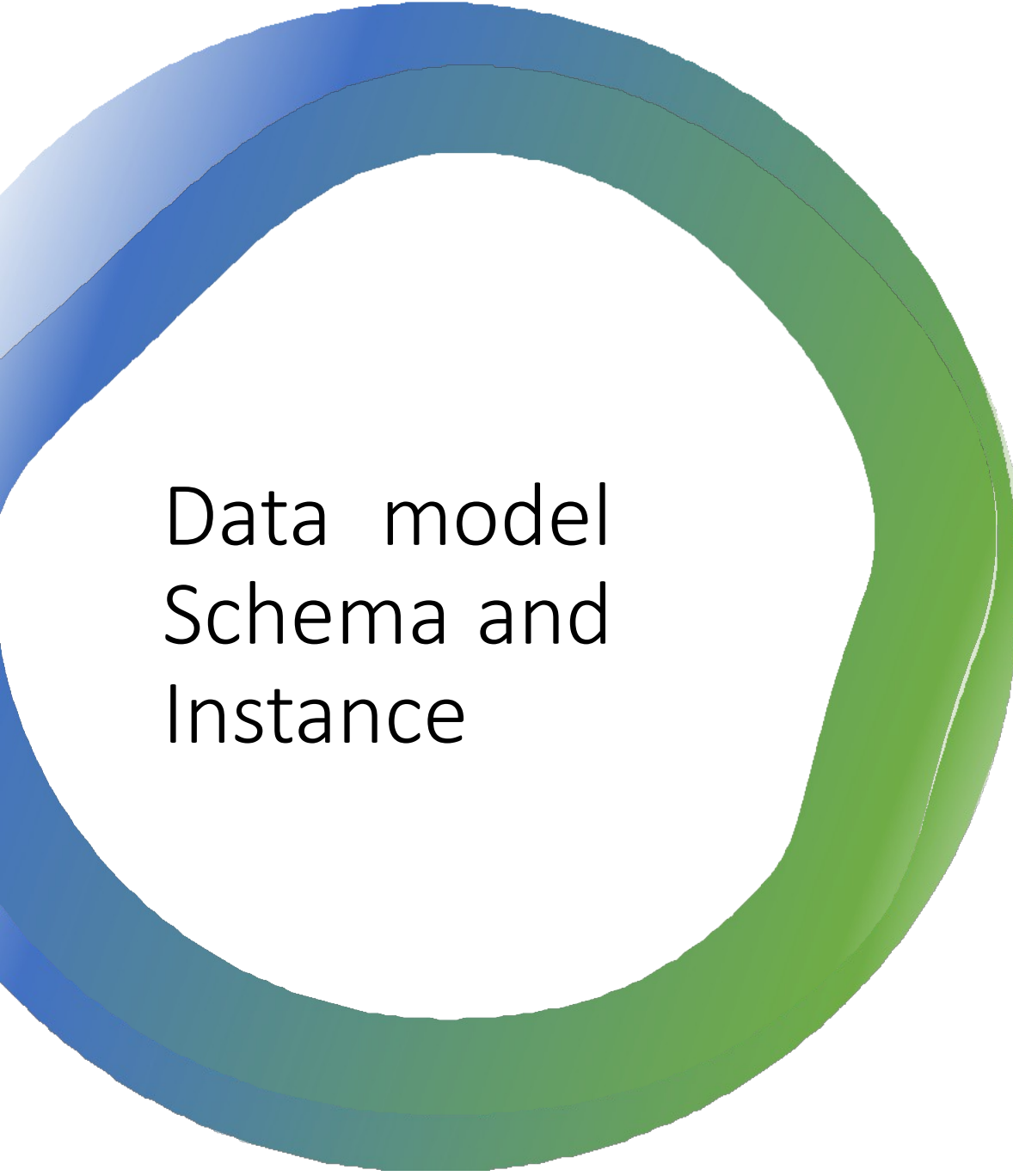
Semi-structured Data Model:

- This type of data model is different from the other three data models (explained above).
- The semi-structured data model allows the data specifications at places where the individual data items of the same type may have different attributes sets.
- The ***Extensible Markup Language***, also known as ***XML***, is widely used for representing the semi-structured data.
- Although XML was initially designed for including the markup information to the text document, it gains importance because of its application in the exchange of data.



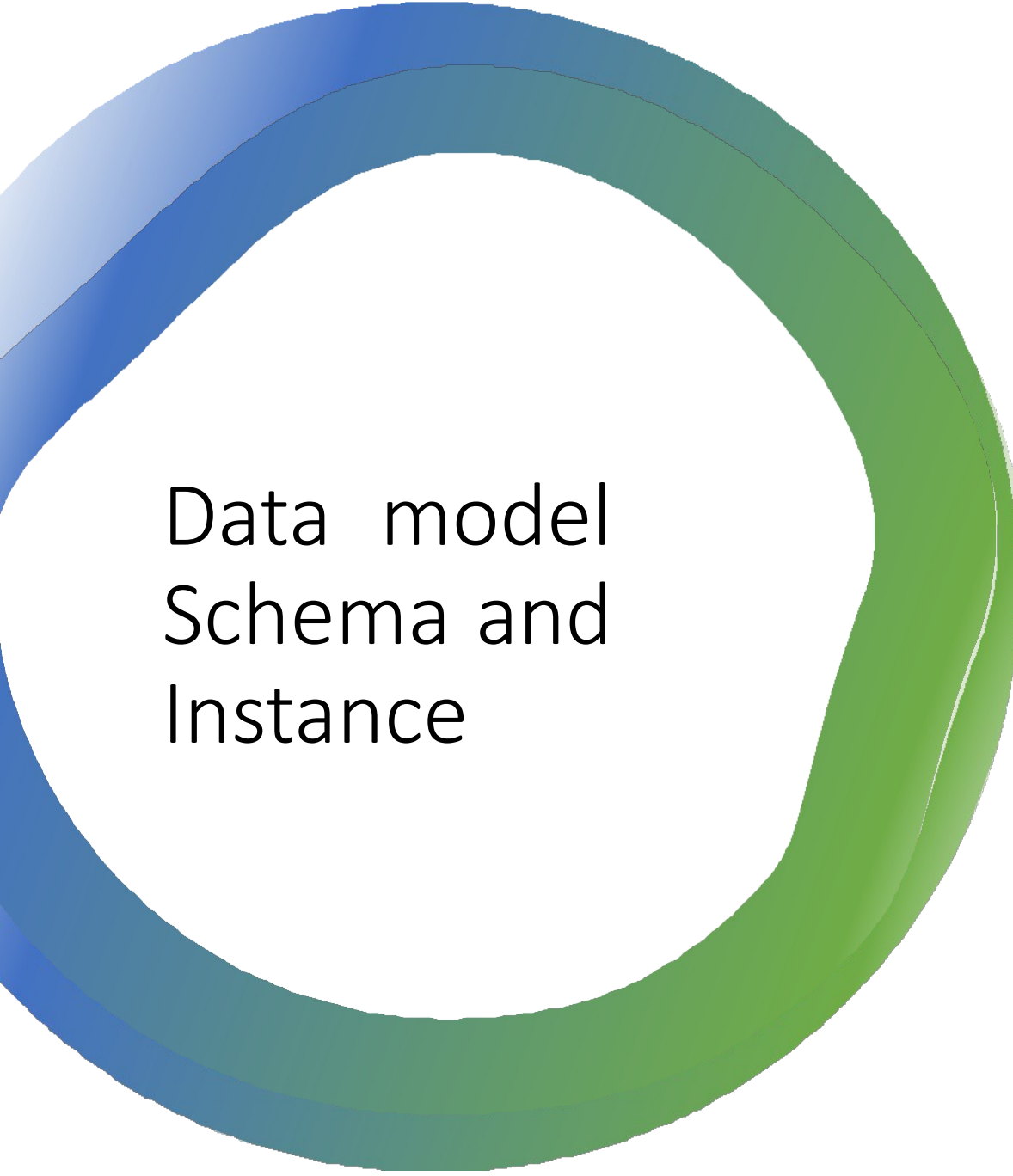
Data model Schema and Instance

- The data which is stored in the database at a particular moment of time is called an **instance** of the database.
- The overall design of a database is called **schema**.
- A database schema is the ***skeleton structure of the database***. It represents the ***logical view*** of the entire database.
- A schema contains schema objects like table, foreign key, primary key, views, columns, data types, stored procedure, etc.
- A database schema can be represented by using the visual diagram. That diagram shows the database objects and relationship with each other.
- A database schema is designed by the database designers to help programmers whose software will interact with the database.
- The process of database creation is called **data modeling**.



Data model Schema and Instance

- A schema diagram can display only some aspects of a schema like the name of record type, data type, and constraints. Other aspects can't be specified through the schema diagram.
 - For example, the given figure (next slide) neither show the data type of each data item nor the relationship among various files.
- In the database, actual data changes quite frequently.
 - For example, in the given figure (Next Slide), the database changes whenever we add a new grade or add a student.
- The data at a particular moment of time is called the ***instance of the database***.



Data model Schema and Instance

STUDENT

Name	Student_number	Class	Major
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COURSE

Course_name	Course_number	Credit_hours	Department
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PREREQUISITE

Course_number	Prerequisite_number
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SECTION

Section_identifier	Course_number	Semester	Year	Instructor
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GRADE_REPORT

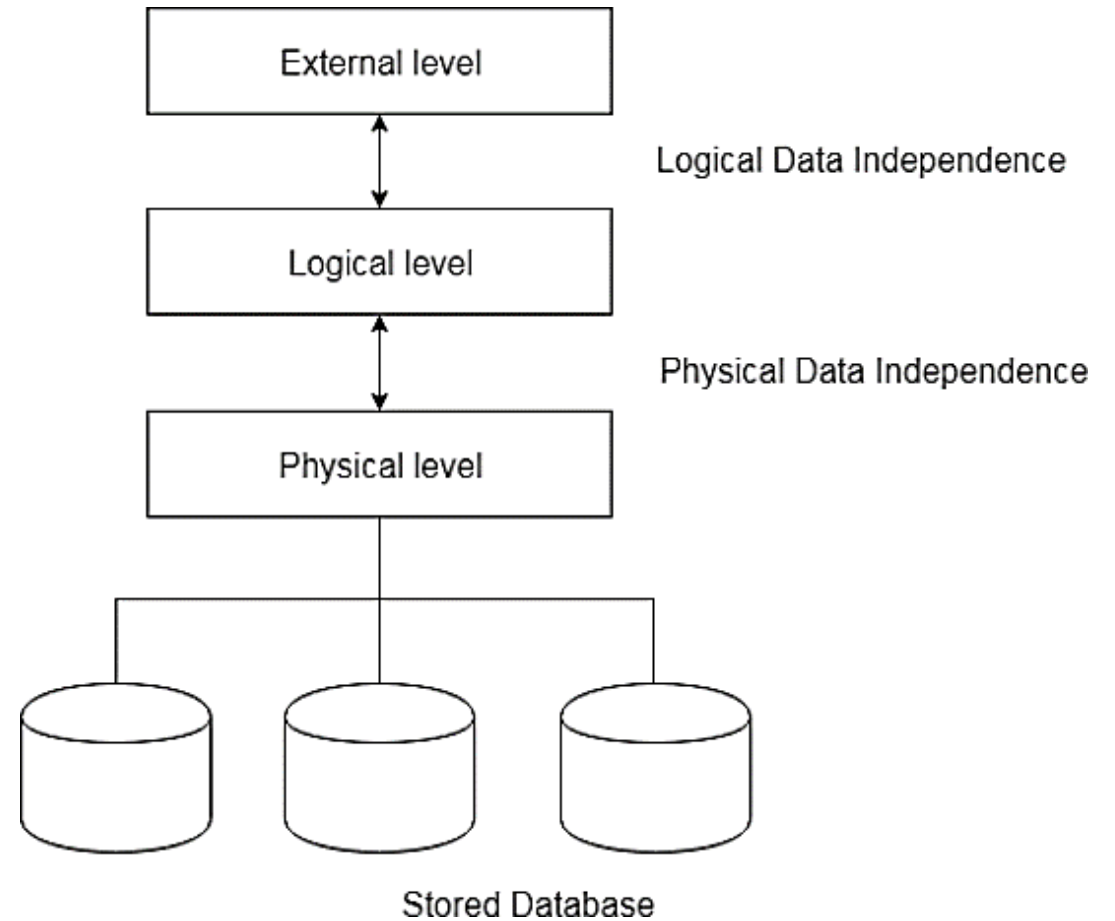
Student_number	Section_identifier	Grade
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Data Independence

- Data independence can be explained using the three-schema architecture.
- Data independence refers characteristic of being able to modify the schema at one level of the database system without altering the schema at the next higher level.
- There are two types of data independence:
 - Logical Data Independence
 - Physical Data Independence

Data Independence





Logical Data Independence

- ***Logical data independence*** refers characteristic of being able to change the conceptual schema without having to change the external schema.
- Logical data independence is used to separate the external level from the conceptual view.
- If we do any changes in the conceptual view of the data, then the user view of the data would not be affected.
- Logical data independence occurs at the user interface level.



Physical Data Independence

- ***Physical data independence*** can be defined as the capacity to change the internal schema without having to change the conceptual schema.
- If we do any changes in the storage size of the database system server, then the Conceptual structure of the database will not be affected.
- Physical data independence is used to separate conceptual levels from the internal levels.
- Physical data independence occurs at the logical interface level.